SIMATS ENGINEERING
ASSESSMENT TOOL 4
COURSE NAME: Artificial Intelligence
COURSE CODE: CSA17

Code of Conduct:

I, **B.Vishnu** / **192512410** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ________________________

Annexure A
Resolution Algorithm Implementation - SOLUTIONS

Question 1 - Rain and Wet Ground

1. Represent the statements in propositional logic:

- Let R = It rains
- Let W = The ground becomes wet
- **Premise 1:** $R \rightarrow W$
- **Premise 2 (Fact):** R
- **Goal:** W

2. Convert all statements into CNF:

- Clause 1 (from Premise 1): $\neg R \lor W$
- Clause 2 (from Premise 2): R

3. Negate the goal:

- Clause 3 (Negated Goal): $\neg W$

4 & 5. Apply the Resolution Algorithm step by step & Show Empty Clause:

- **Step 1:** Resolve Clause 1 ($\neg R \lor W$) with Clause 2 (R).
- *Derived Clause 4: W*
- **Step 2:** Resolve Derived Clause 4 (W) with Clause 3 ($\neg W$).
- *Result:* Empty Clause ($\varnothing$)
- **Conclusion:** Since the empty clause is derived, a contradiction is found. The original goal that the ground is wet (W) is proved.

Question 2 - Student Assignment Submission

1. Write the propositional logic expressions:

- Let S = Rahul submitted the assignment
- Let I = Rahul receives internal marks
- **Premise 1:** $S \rightarrow I$
- **Premise 2 (Fact):** S
- **Goal:** I

2. Convert the premises into CNF:

- Clause 1: $\neg S \lor I$
- Clause 2: S

3. Negate the goal:

- Clause 3: $\neg I$

4 & 5. Apply the Resolution Algorithm & Conclude:

- **Step 1:** Resolve Clause 1 ($\neg S \lor I$) with Clause 2 (S).
- *Derived Clause 4: I*
- **Step 2:** Resolve Derived Clause 4 (I) with Clause 3 ($\neg I$).
- *Result:* Empty Clause ($\varnothing$)
- **Conclusion:** The empty clause is successfully derived. Therefore, the goal is proved: Rahul receives internal marks.

Question 3 - Library Membership

1. Represent the knowledge base using propositional logic:

- Let M = Priya is a library member
- Let B = Priya can borrow books
- **Premise 1:** $M \rightarrow B$
- **Premise 2 (Fact):** M
- **Goal:** B

2. Convert each statement into CNF:

- Clause 1: $\neg M \lor B$
- Clause 2: M

3. Negate the goal:

- Clause 3: $\neg B$

4 & 5. Perform resolution & State conclusion:

- **Step 1:** Resolve Clause 1 ($\neg M \lor B$) with Clause 2 (M).
- *Derived Clause 4: B*
- **Step 2:** Resolve Derived Clause 4 (B) with Clause 3 ($\neg B$).
- *Result:* Empty Clause ($\varnothing$)
- **Conclusion:** Contradiction reached. The original goal is proved: Priya can borrow books.

Question 4 - Placement Eligibility

1. Convert the statements into propositional logic:

- Let C = Arun cleared the aptitude test
- Let E = Arun is eligible for placement
- **Premise 1:** $C \rightarrow E$
- **Premise 2 (Fact):** C
- **Goal:** E

2. Write the CNF clauses:

- Clause 1: $\neg C \lor E$
- Clause 2: C

3. Negate the goal:

- Clause 3: $\neg E$

4 & 5. Apply the Resolution Algorithm step by step & Draw tree/conclude:

- **Step 1:** Resolve Clause 1 ($\neg C \lor E$) with Clause 2 (C).
- *Derived Clause 4: E*
- **Step 2:** Resolve Derived Clause 4 (E) with Clause 3 ($\neg E$).
- *Result:* Empty Clause ($\varnothing$)
- **Conclusion:** Resolving the knowledge base with the negated goal results in the empty clause. We conclude that Arun is eligible for placement.

Question 5 - Access Control System

1. Represent the knowledge base in propositional logic:

- Let P = User enters the correct password
- Let A = User is authenticated
- Let G = User is granted access
- **Premise 1:** $P \rightarrow A$
- **Premise 2:** $A \rightarrow G$
- **Premise 3 (Fact):** P
- **Goal:** G

2. Convert all implications into CNF:

- Clause 1: $\neg P \lor A$
- Clause 2: $\neg A \lor G$
- Clause 3: P

3. Negate the goal:

- Clause 4: $\neg G$

4 & 5. Apply Resolution to derive Empty Clause & Explain:

- **Step 1:** Resolve Clause 1 ($\neg P \lor A$) with Clause 3 (P) by canceling the complementary literals $\neg P$ and P.
- *Derived Clause 5: A*
- **Step 2:** Resolve Clause 2 ($\neg A \lor G$) with Derived Clause 5 (A) by canceling $\neg A$ and A.
- *Derived Clause 6: G*
- **Step 3:** Resolve Derived Clause 6 (G) with Clause 4 ($\neg G$) by canceling G and $\neg G$.
- *Result:* Empty Clause ($\varnothing$)
- **Final Conclusion:** Because an empty clause (contradiction) was reached, the negated goal is false. The original goal is thus valid, and the user is granted access.